A/B Test Plan

Element being changed: I will be changing the ***discount-sports-equipment.html*** page because it is the highest-volume query in my keyword plan, so people that are searching for it are driven by price. The specific element that will be changed is the opening paragraph, which is the first thing a visitor sees when they look under the headline. Nothing else will change, and both versions have the same headline, image, table, links, and are about 40 to 45 words.

Version A (current page): "Used gear is already well below retail, but we also run regular markdowns so you can find cheap sports equipment at even better prices. Here is how our discounts work and when to shop for the best deals."

Version B: "Items on our clearance rack drop 25 percent after 45 days and 50 percent after 90 days. Starter bundles take 15 percent off, and monthly markdowns reach as high as 45 percent. Here is how each discount works and when to shop for the best deals."

Hypothesis: If the opening paragraph mentions the specific discount amounts instead of describing the discount in general terms, then a larger number of AI visitors playing a price-conscious person will say they would contact Second Wind, because concrete saving is easier for a shopper to judge than a generic promise.

Model, Persona, Prompt, Assignment: The model that will be used is Claude Sonnet 5.5, with the same settings for every trial and memory off.

Prompt: "You are a parent in Iowa City on a tight budget, shopping for discounted sports equipment for your child's upcoming season. You have already looked at two other shops and will only contact one, so you are looking for a reason to pick this one. Treat the business below as real. Read the page and decide whether you would contact this business (call, email, or visit the shop). Answer yes or no, then give one sentence of reasoning."

I will generate 120 assignments at once in a spreadsheet column using =IF(RANDBETWEEN(0,1)=0,"A","B"), which will act as a coin flip for each trial. Then I will copy those results as fixed values so they do not change and run the trials in row order.

Outcome: The primary outcome is if the model/persona would contact this business via call, email, or visiting the shop. An answer that has a Yes will be measured as 1, and anything else such as No, Maybe, or other answers, will be measured as 0. The full output will be kept for every trial and no trial is duplicated.

Sample Size and Date: The sample size will be 120 trials in total, with about 60 per version. I will only stop when I have run all 120 trials and not look at any Yes or No outcomes until the end. I expect to reach the total sample size by around November 15 or 16, before the latest allowed launch date.

Comparison and Decision: I will compare the share of Yes answers in Version B to the share of Yes answers in Version A using a z test. I will report the difference, including the standard error, z score, p-value, and a 95 percent confidence interval. The significance level will be set at 0.05. The result that would count as support for my hypothesis would be Version B's yes share is higher than Version A's and if $p < 0.05$. I will only use Version B if I see that the hypothesis is supported.